

9.3.4 HSI48 clock

The HSI48 clock signal is generated from an internal 48 MHz RC oscillator and can be used directly for USB and for random number generator (RNG) as well as SDMMC.

The internal 48 MHz RC oscillator is mainly dedicated to provide a high-precision clock to the USB peripheral by means of a special clock recovery system (CRS) circuitry. The CRS can use the USB SOF signal, the LSE or an external signal to automatically and quickly adjust the oscillator frequency on-fly. It is disabled as soon as the system enters Stop or Standby mode. When the CRS is not used, the HSI48 RC oscillator runs on its default frequency which is subject to manufacturing process variations.

For more details on how to configure and use the CRS peripheral please refer to [Section 10: Clock recovery system \(CRS\)](#).

The HSI48RDY flag in the RCC_CRRCR register indicates whether the HSI48 RC oscillator is stable or not. At startup, the HSI48 RC oscillator output clock is not released until this bit is set by hardware.

The HSI48 can be switched on and off using the HSI48ON bit in the RCC_CRRCR register.

9.3.5 PLL

The device embeds three PLLs: PLL, PLLSAI1 and PLLSAI2. Each PLL provides up to three independent outputs. The internal PLLs can be used to multiply the HSI16, HSE or MSI output clock frequency. The PLLs input frequency must be between 4 and 16 MHz. The selected clock source for each PLL is divided by a dedicated programmable factor PLLM, PLLSAI1M, PLLSAI2M, from 1 to 8 to provide a clock frequency in the requested input range. Refer to [Figure 30: Clock tree](#), [RCC PLL configuration register \(RCC_PLLCFGR\)](#), [RCC PLLSAI1 configuration register \(RCC_PLLSAI1CFGR\)](#) and [RCC PLLSAI2 configuration register \(RCC_PLLSAI2CFGR\)](#).

The PLLs configuration (selection of the input clock and multiplication factor) must be done before enabling the PLL. Once the PLL is enabled, these parameters cannot be changed.

To modify the PLL configuration, proceed as follows:

1. Disable the PLL by clearing PLLON to 0 in [RCC clock control register \(RCC_CR\)](#).
2. Wait until PLLRDY bit is cleared. The PLL is now fully stopped.
3. Change the desired parameter.
4. Enable the PLL again by setting PLLON bit to 1.
5. Enable the desired PLL outputs by configuring PLLPEN, PLLQEN, PLLREN bits in [RCC PLL configuration register \(RCC_PLLCFGR\)](#).

An interrupt can be generated when the PLL is ready, if enabled in the [RCC clock interrupt enable register \(RCC_CIER\)](#).